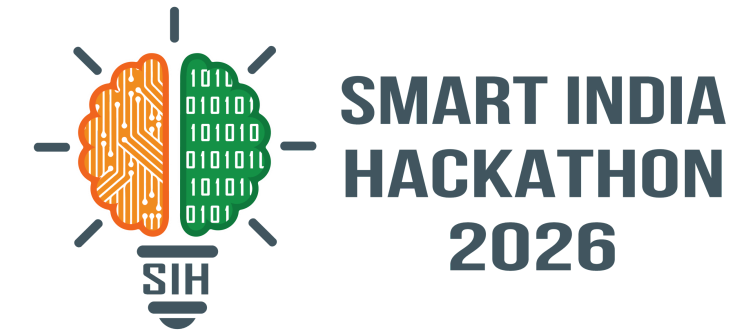
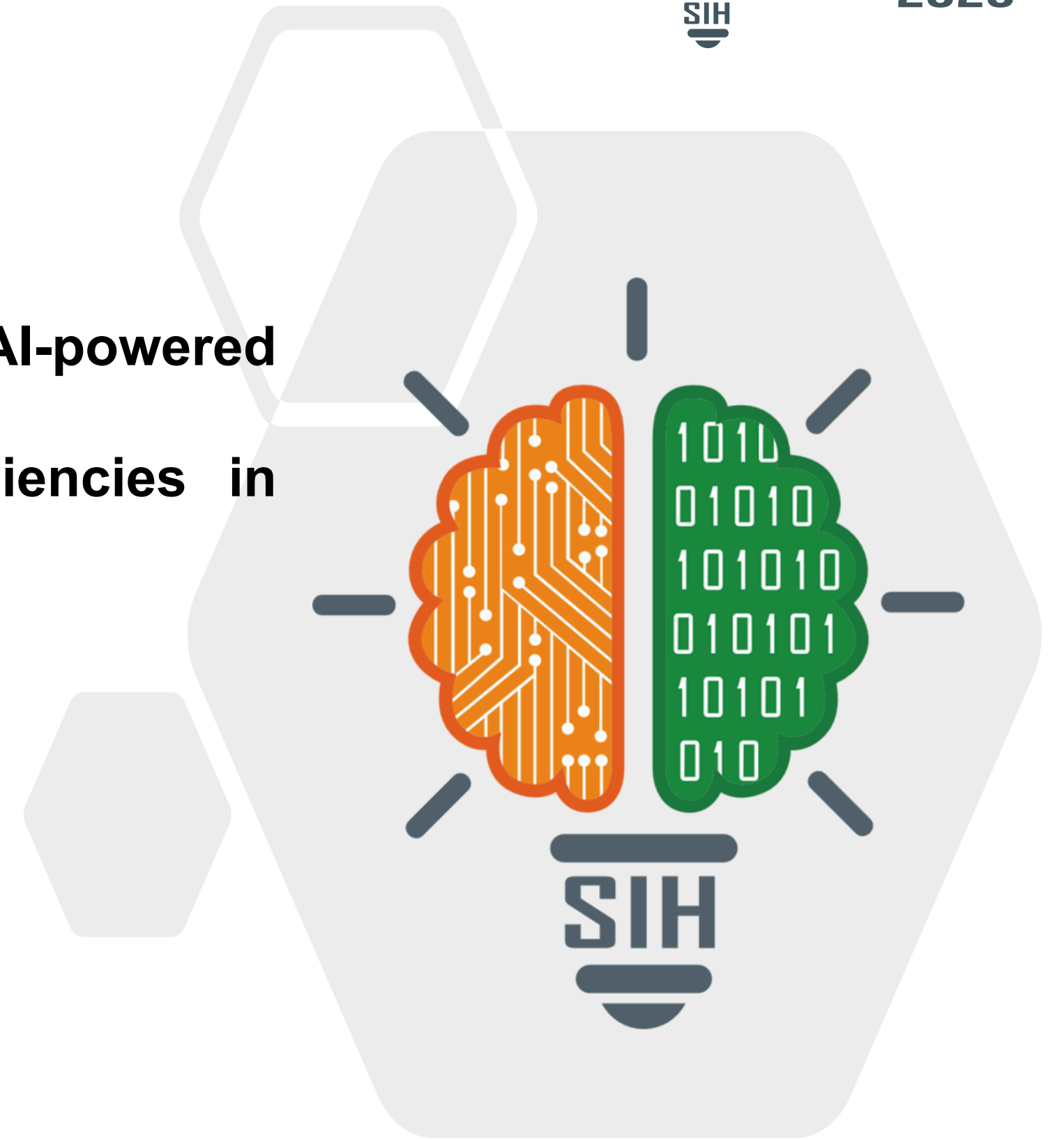


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26102**
- **Problem Statement Title- Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID- 133267**
- **Team Name - MetaMinds_404**



Proposed Solution - 5 main components

Government Intelligence Web Portal

- AI-based risk scoring & anomaly detection
- Progress-expenditure analysis
- Contractor risk intelligence
- AI-recommended corrective actions
- Charts, statistics & project analytics
- RAG-powered multilingual chatbot/VA

Citizen Participation & Transparency

- Recommend new MPLADS works
- Location-tagged citizen photographs
- Search & track projects
- View permitted project information
- Submit grievances & feedback
- Multilingual AI assistance

Trusted Evidence & Audit Layer

- Tamper-evident evidence storage
- CID-linked inspection records
- Permissioned consortium ledger
- RBAC-based approvals
- Verifiable project event history

Smart-Contract Escrow & Fund Control

- Project-level escrow fund tracking
- Smart-contract-based release rules
- Automatic fund hold/freeze for high-risk or non-compliant projects
- Milestone & approval-based disbursement
- Authorized human override

Field Officer Mobile App

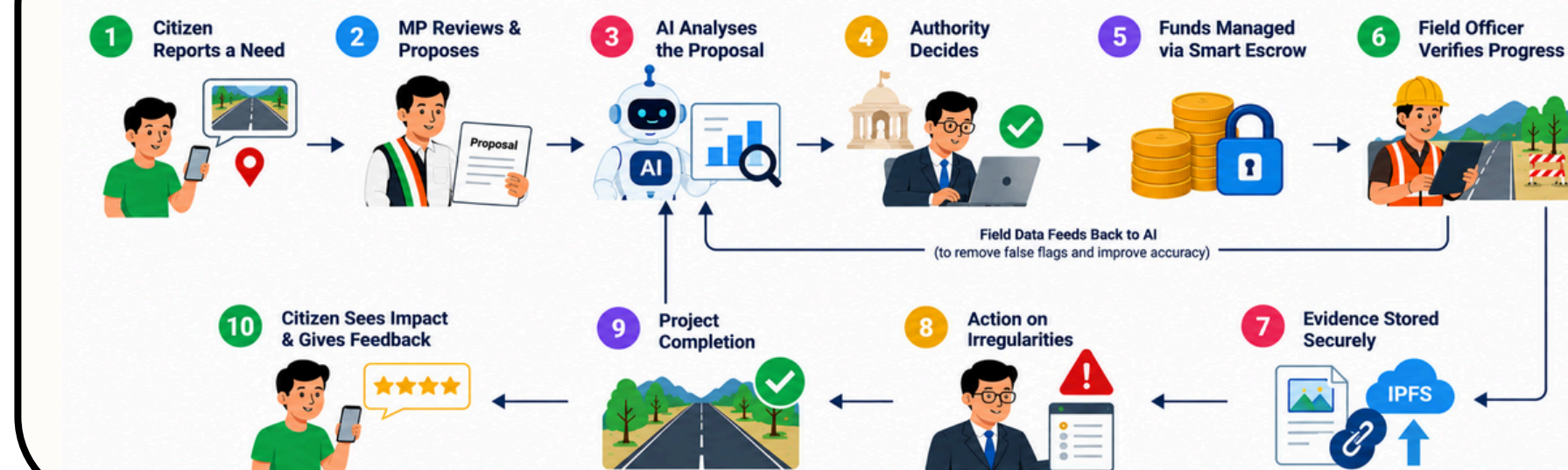
- Assigned project inspections
- GPS & timestamp verified photographs
- Direct IPFS evidence upload
- Digital sign-off
- Human-in-the-loop verification



REQUIREMENT OF THE PS: Anomalies & Fraud Detection

OUR SOLUTION: Detection + Prevention + Site Verification + Web3 Tamper proof Audit Compliance

User Flow Diagram



Innovation and Uniqueness

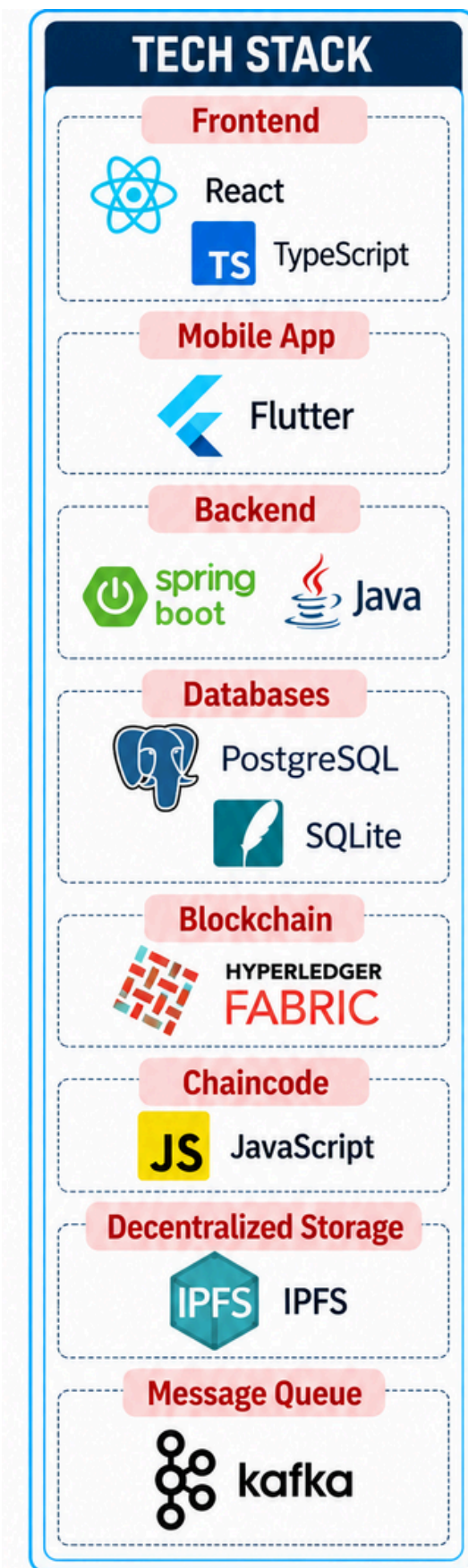
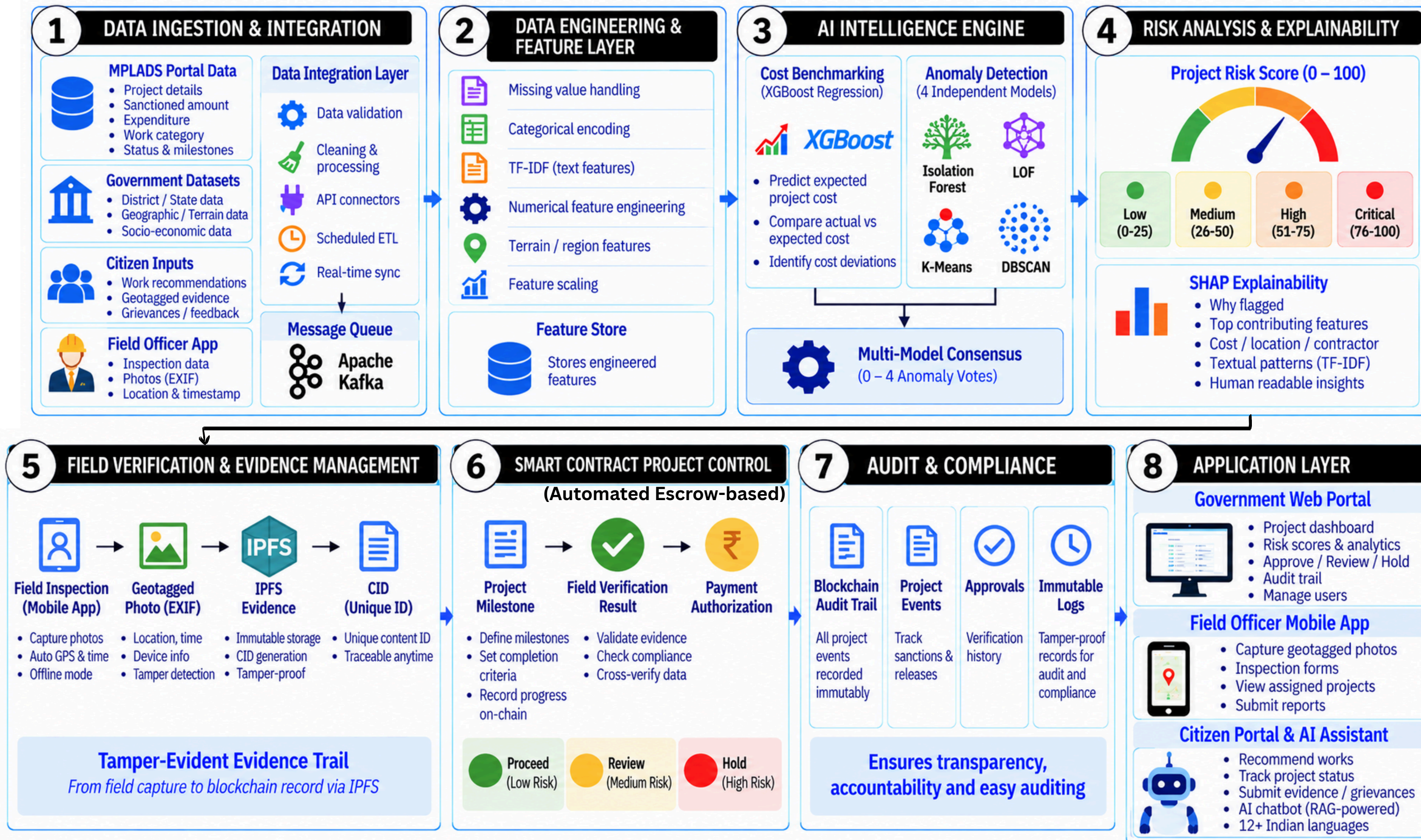
- **Image Verification:** Direct image upload to **IPFS** provides verifiable image provenance, **eliminating** need for separate **SHA-256, OpenCV, or dHash-based image forensics**.
- **Smart-Contract Fund Control:** **Automated escrow-based** fund monitoring and risk-triggered **fund freezing**.
- **Permissioned Web3 Governance:** **Consortium Hyperledger Fabric** with **RBAC** and **digitally signed project** events.
- **AI Consensus Risk Engine:** **DBSCAN, K-Means, Isolation Forest, LOF & XGBoost** for multi-dimensional risk detection, with **SHAP** for explainable and actionable risk insights.
- **Prevention along with cure:** A **newly proposed and unsanctioned project** will also be **scored** based on contractors, MP history, suspicious proposal metrics **to prevent fraud beforehand**.
- **Terrain-Aware & Graph-Based Intelligence:** Regional and terrain factors **reduce geographical bias**, while relationship analysis identifies contractor-work-authority risk patterns.
- **Verifiable Evidence Provenance:** **EXIF-verified** field evidence stored on **IPFS** through **CID-based** traceability, alongside citizen **location-tagged evidence**.
- **Citizen-Powered Ground Intelligence:** Citizens can submit and track **grievances** through **Telegram, WhatsApp**, with **guided conversations**, ticket IDs and status tracking.

How it addresses the problem?

- **Limited Citizen Transparency:** Citizen portal provides permitted project, **progress** and **expenditure visibility**.
- **Slow Decision-Making:** **SHAP explanations** + AI-recommended actions help authorities prioritize interventions.
- **Language & Accessibility Barriers:** **RAG-powered** multilingual AI Voice assistant & Chatbot enables access in **12+ languages**.
- **Financial Fraud & Misuse:** AI anomaly detection + cost benchmarking + smart-contract fund controls.
- **Fragmented Monitoring:** Unified project, financial, contractor and field data enables centralized monitoring.
- **Verification Gaps:** **GPS/EXIF-based** field inspections provide traceable **ground-level evidence**.
- **Evidence Tampering and Integrity Risk:** **IPFS CIDs + permissioned blockchain** create verifiable evidence provenance.
- **Dependence on Foreign AI:** **Sovereign AI** with **Sarvam AI** reduces reliance on **foreign AI** platforms and supports **greater control** over **sensitive governance data**, while enabling **multilingual** citizen **interaction**.



TECHNICAL APPROACH



FEASIBILITY

What is CAG?
Audits government accounts and checks how public money is spent.

CHALLENGES & SOLUTIONS

Technical Feasibility

MPLADS is fully digital via e-SAKSHI. Our prototype already risk-scores **6,000+** real works — **CAG (Comptroller and Auditor General of India)** once found **₹73.76 Cr** spent on **2,340** non-compliant works. **Real-Time Data Ingestion** refreshes 100% real MPLADS records on a configurable schedule, so nothing shown is stale or fabricated.

Financial Feasibility

Zero licensing cost open-source stack, deployable on India's own **NIC/MeghRaj cloud**, and built to ride the **₹21,936 Cr** MeitY Digital India push (2024-25, +52% YoY), so it fits within existing digital budgets.

Operational Feasibility

With **1.09 billion** internet users and rural access up **10% → 48%** since 2015, citizens can now raise grievances and recommend works online, while our **offline-first Flutter app** captures GPS/EXIF-verified field evidence and syncs once connected.

Regulatory Feasibility

CAG flagged **₹1,788 Cr** unspent, **₹53.74 Cr** on inadmissible works, and 775 works (**₹10.18 Cr**) never taken up, plus 568 delayed works (**₹7.30 Cr**). Our tool **auto-flags dormant, delayed and duplicate works** for auditors, with no new legislation.

VIABILITY

Market Viability

India's e-governance market is projected to grow from **\$1.69B** (2025) to **\$5.5B** by 2035 at a **12.5%** CAGR — the same Digital India push already funding tools like ours.

Sustainable Viability

MPLADS is a permanent, recurring scheme (**₹862.97 Cr** in FY26 works alone) — the same architecture extends to sibling schemes like MLALADS and District Mineral Foundation funds.

1. Integrating a new oversight layer across MoSPI, State, District, MP and Citizen roles without disrupting existing MPLADS workflows

SOLUTION



Regional language access

- Multilingual portal + voice assistant support



Citizen transparency without data leaks

- Server-narrowed public API + IPFS-backed grievance photos with tracked status

2. Verifying that a work's claimed on-ground progress and evidence are genuine, not fabricated or reused

SOLUTION



Catch Faked location claims

- Device GPS + EXIF cross-checked against site



Reused/duplicate/deepfake photos detection

- IPFS CID match, instead of dHash and SHA-256 for decentralized cloud storage as well.



Tackling a single corrupt approver flow

- Dual-authority sign-off before finalizing

3. Keeping the audit trail tamper-proof and catching repeat bad actors, without full blockchain infrastructure on day one

SOLUTION



Tamper-proof event history

- Blockchain-based Immutable audit trail using Hyperledger Fabric anchoring



Flag Repeat bad actors & notify unnoticed behavior

- Auto-derived red-flag registry, not a manual list & Automated milestone tracker + alert notifications



Watching the watchers

- Officer override-rate & turnaround-time dashboard

Metrics for checking credibility of an MP/Project



- Risk Score** auto-flags a project Low/Medium/High using dormancy, missing payments & cost anomalies



- Fund Utilisation** % exposes whether an MP's sanctioned money is actually being spent and delivered



- Vendor Concentration** catches favoritism when one contractor wins a suspicious share of an MP's works



- Duplicate Work Detection** flags the same project sanctioned or listed more than once

- Peer Benchmarking** ranks an MP against others on utilisation, delays & completion rate

IMPACT AND BENEFITS

IMPACTS AS PER STAKEHOLDERS

Government & MPs

- **Data-Driven Governance:** AI risk intelligence enables faster, evidence-based intervention.
- **Financial Oversight:** Automated fund controls and alerts strengthen accountability.

Field Officers & Agencies

- **Efficient Verification:** GPS-verified mobile inspections simplify ground-level monitoring.
- **Traceable :** Digital evidence and sign-offs improve implementation accountability.

Citizens

- **Greater Transparency:** Access project progress, expenditure and permitted public records.
- **Active Participation:** Recommend works and submit location-tagged ground evidence.

Public & Society

- **Better Public Spending:** Early detection of anomalies, delays and cost inefficiencies.
- **Trustworthy Development:** Verifiable evidence and auditable project histories strengthen public trust.



BENEFITS OF THE SOLUTION:

- **For Economy:** Reduces financial leakage and improves public fund utilization.
- **For Governance:** Enables proactive, data-driven monitoring and accountability.
- **For Society:** Promotes inclusive citizen participation and transparent development.
- **For Infrastructure:** Improves project delivery, verification and resource efficiency.
- **For Public Trust:** Creates verifiable evidence and traceable project histories.

We use **Sovereign AI**: Our RAG-powered chatbot & voice assistant uses **Sarvam AI** to enable citizens to interact in their own **Indian languages**, promoting **Make in India** through **indigenous AI technology**.

SUSTAINABLE DEVELOPMENT GOALS (SDGS) OF THE SOLUTION - AS PER THE UNITED NATIONS ARE:



• **SDG 9** — Industry, Innovation & Infrastructure



• **SDG 11** — Sustainable Cities & Communities



• **SDG 16** — Peace, Justice & Strong Institutions



• **SDG 17** — Partnerships for the Goals

UNIQUE SELLING POINTS (USPs) & NOVELTIES

TECHINICAL USP_s

USP 1

Blockchain-Powered Financial & Audit Governance

- **Smart-contract escrow:** rule-based fund release, milestone validation & **automatic fund freeze** for **high-risk/non-compliant projects**
- **Permissioned consortium blockchain:** Hyperledger Fabric + RBAC + digital signatures
- Tamper-evident project event & approval history across authorized stakeholders

USP 2

Trusted Geospatial Evidence & Provenance

- **EXIF-based geotagging:** GPS coordinates, timestamp & officer/work identity embedded into field photographs
- **IPFS/Pinata evidence storage:** content-addressed CIDs for verifiable evidence provenance
- Separate evidence handling for **officer inspections and citizen-submitted recommendations**

USP 3

Multi-Model AI Consensus & Geographical Fairness

- **Ensemble anomaly detection:** **DBSCAN + K-Means + Isolation Forest + LOF**
- **XGBoost cost benchmarking:** expected vs. actual project expenditure
- **Terrain/region-aware normalization:** reduces geographic bias in risk scoring
- **Consensus across models** produces a more robust project risk score

FEATURE WISE USP_s

USP 4

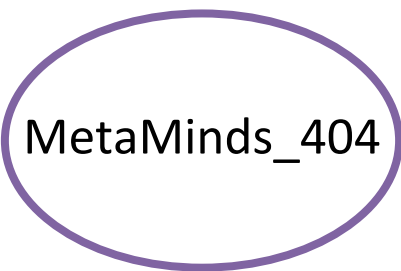
Explainable & Action-Oriented AI

- **SHAP-powered explainability:** identifies the factors driving each risk score
- AI generates **recommended corrective actions** based on project risk and detected anomalies
- **High-risk contractor identification** → risk tagging / blacklist intelligence
- **Automated email alerts & notifications** to relevant authorities/MPs

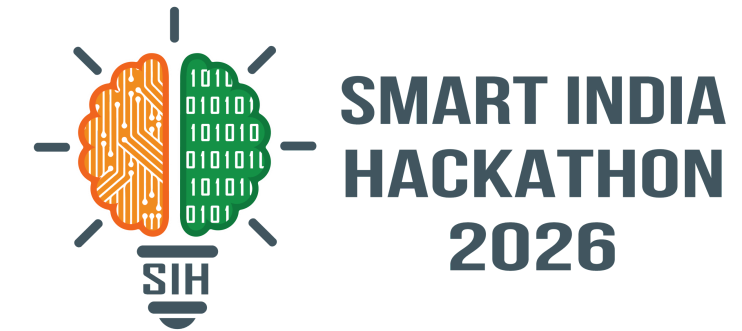
USP 5

Citizen + Intelligent Access Layer

- **Citizen participation:** **Recommend works**, submit location-tagged **photographic evidence**, and **raise grievances** through **WhatsApp & Telegram** with ticket-based tracking.
- **RBAC-based ecosystem:** Government / Officer / Citizen access with **role-specific capabilities**
- **RAG-powered AI Voice assistant & chatbot**
- **12+ language multilingual support** across citizen and government interfaces
- **Interactive charts, statistics** & visual analytics.



RESEARCH AND REFERENCES



Images of prototype (portal, app, bot)

1. MPLADS eSAKSHI Public Dashboard

mplads.mospi.gov.in/digigov/dashboard.html

The official MPLADS dashboard provides project, expenditure and implementation information and forms the primary data source for our proposed monitoring system.

2. MPLADS – Official Portal

mplads.gov.in

The portal provides MPLADS guidelines, work information and monitoring-related resources, helping understand the existing implementation framework.

3. Revamped Public Dashboard of MPLADS eSAKSHI Portal – PIB

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2235932&lang=1®=3>

4. AI for Public Expenditure Analysis

<https://www.sciencedirect.com/science/article/pii/S1877050925035513> ScienceDirect

The research explores AI and machine learning for anomaly identification, financial auditing, expenditure efficiency and fiscal accountability in the public sector.

5. Blockchain-Based Transparency in Government Records

<https://www.sciencedirect.com/science/article/pii/S2667096824000934>

The study demonstrates how blockchain-based government records can improve transparency, accountability and verifiable audit trails.

6. Blockchain Governance in the Public Sector

<https://www.sciencedirect.com/science/article/pii/S0740624X21000617>

The research examines blockchain adoption in public-sector environments, including architecture, interoperability, accountability and governance considerations.